

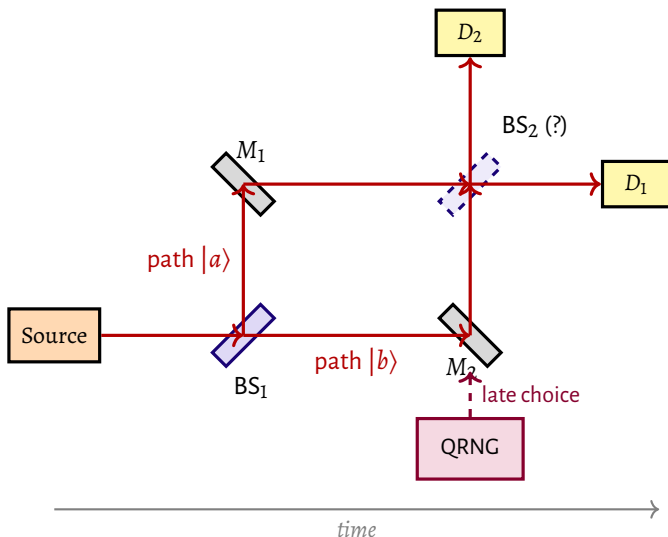


Figure 12 – **Schematic of the delayed-choice experiment (Mach-Zehnder version).** A photon emitted by the source strikes the beam splitter BS_1 , creating a coherent superposition of the paths $|a\rangle$ (via mirror M_1) and $|b\rangle$ (via mirror M_2). **Crucial decision:** after the photon has already passed through BS_1 and is in the arms of the interferometer, a quantum random number generator (QRNG) decides whether the second splitter BS_2 will be present or absent. If BS_2 is *present*, the paths recombine and the photon exhibits *interference* (wave behavior) — detectors D_1 and D_2 register counts with probabilities depending on the relative phase between the arms. If BS_2 is *absent*, each detector is connected uniquely to one of the paths, revealing a *definite trajectory* (particle behavior). Wheeler’s question was: how does the photon “know” which behavior to adopt if the decision was made after it had already entered the interferometer?

